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S.C. Dwivedi

LED STRIP DIMMER And CALL BELL

SURESH CHANDRA DWIVEDI

An LED dimmer circuit adjusts the brightness of LED strips, serving both as an energy-saving tool and a method to create aesthetic lighting. Call bells, designed to draw attention, are commonly used in various environments such as offices, hospitals and residential care homes. Here, we explore a simple 12V LED dimmer circuit, incorporating an LM556 dual timer IC, which also functions as a call bell. The LM556 dual timer IC houses two 555 timers within a single 14-pin package. In this LM556 PWM-based LED dimmer circuit, the brightness of the LEDs is effortlessly controlled using a potentiometer. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the LED strip dimmer and call bell, as shown in Fig. 2, is built around a 230V AC primary to 15V, 500mA secondary transformer (X1), a bridge rectifier (BR1), a 12V voltage regulator (LM7812, IC1), a dual timer (LM556, IC2), a MOSFET (IRFZ44, T1), a transistor (BC337, T2), an 8-ohm 0.5-watt speaker (LS1), and a few other components.

The 230V AC mains voltage is stepped down to 15V through step-

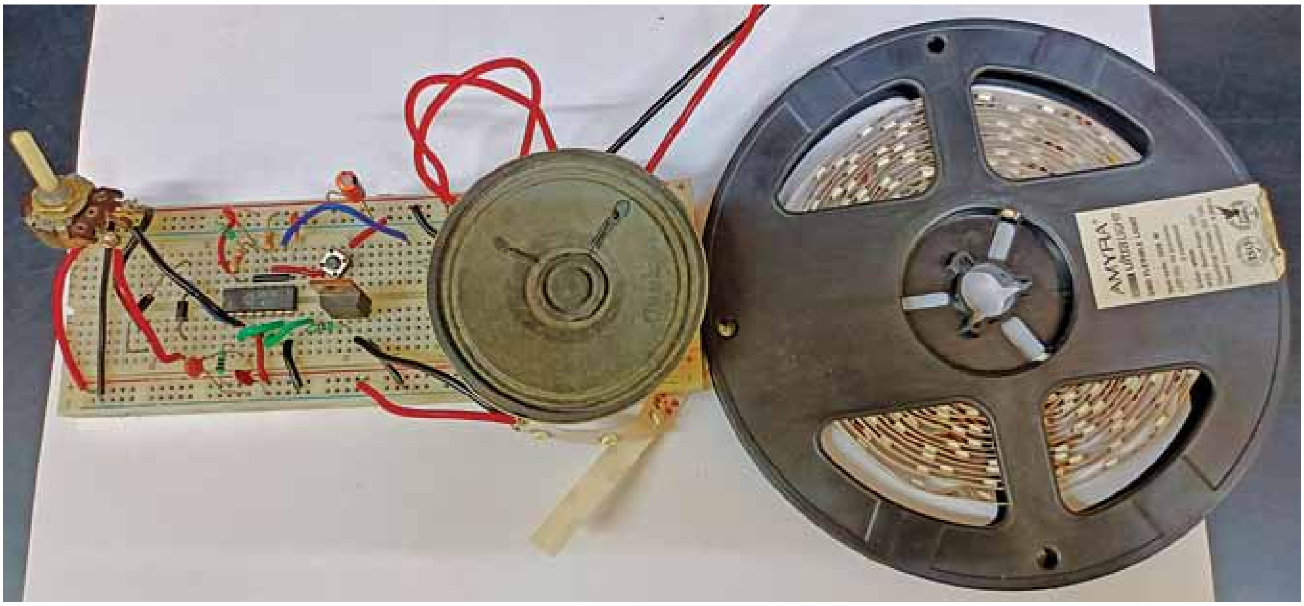

Fig. 1: Author's prototype tested on a breadboard in EFY Lab

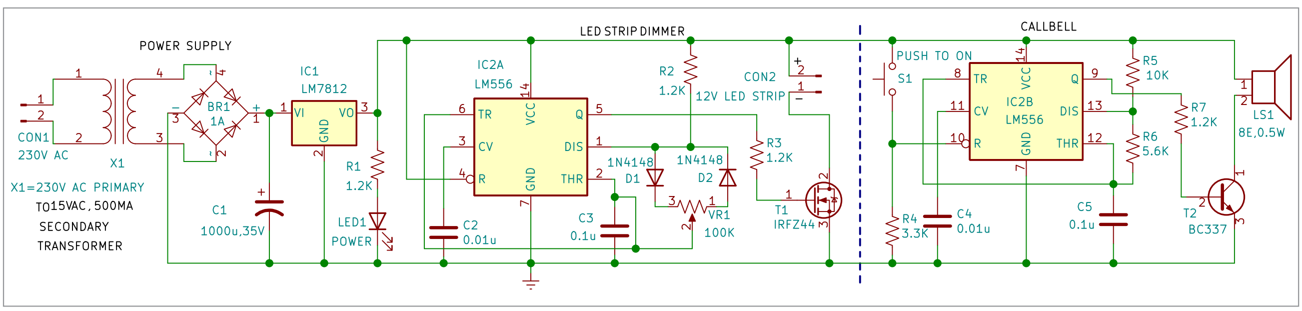

Fig. 2: Circuit diagram